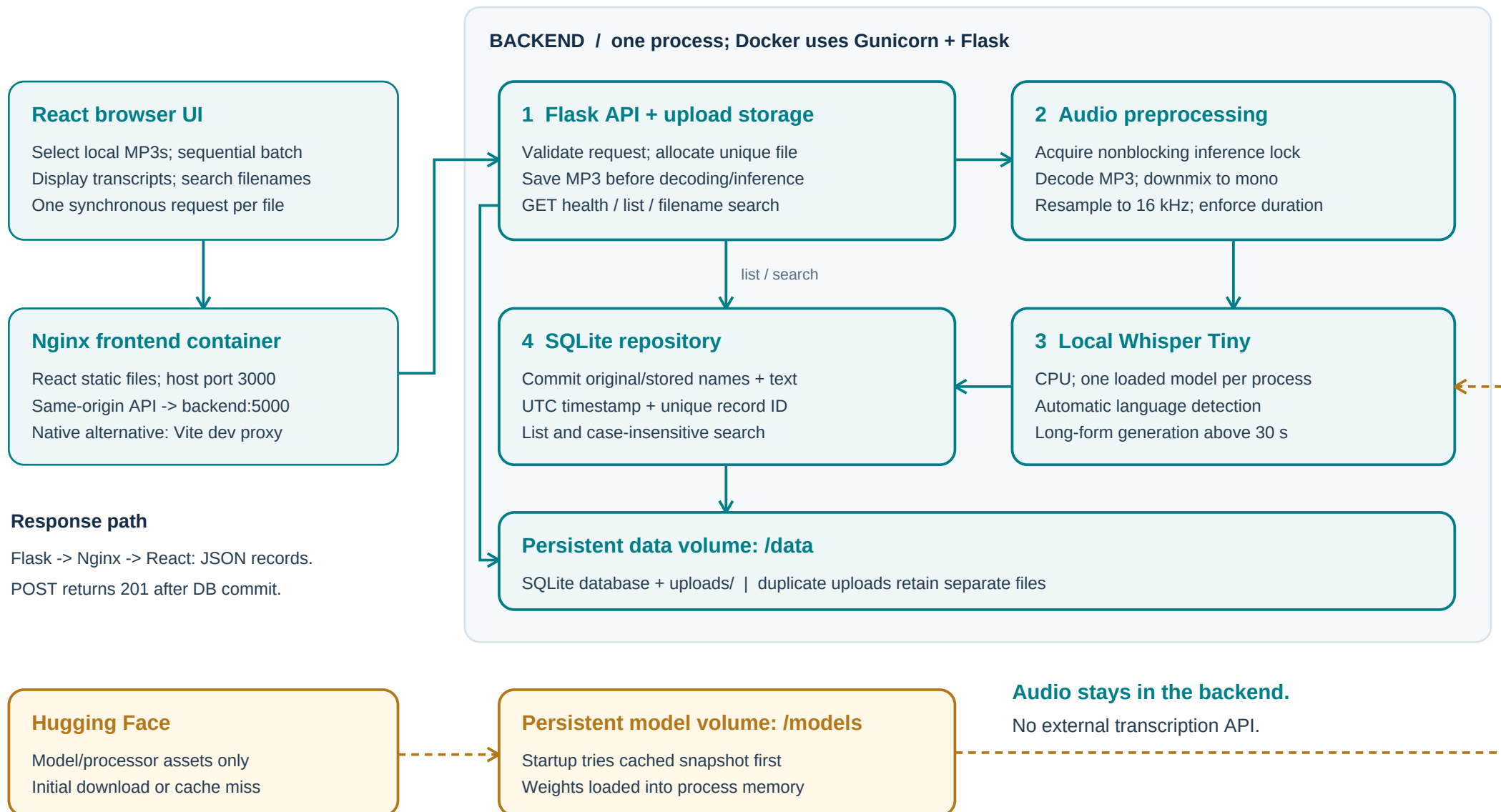


Architecture and data flow

Solid arrows: runtime work and data. Dashed amber arrows: model provisioning and startup.



Assumptions, operations and evidence

01 Request and storage behavior

MP3 only; 25 MiB full-request limit; 600 s decoded audio limit.
React sends batch items sequentially. Overlapping inference returns 503 service_busy; no unbounded inference queue.
Files use sanitized unique names and exclusive creation.
Decode/inference/DB failures remove the saved upload.
SQLite and filesystem are not jointly atomic: an abrupt crash can leave an orphan file. No crash reconciliation is implemented.

03 Startup and model boundaries

Hugging Face supplies openai/whisper-tiny model assets.
A complete local cache avoids repeated weights downloads.
The loaded model is shared inside one backend process.
GET /health checks model and database readiness; model-load failure yields 503. Restart after correcting configuration/network.
Health is not a guarantee that a future write will succeed.
Timeouts are provisional; threaded worker timeout is not a hard per-request deadline. Longer workloads need measurement.

02 Deployment assumptions

One backend replica, Linux/x86-64, CPU inference.
Gunicorn: one gthread worker, four threads; UID/GID 10001.
Nginx serves React on host 3000; API also on host 8000.
Data and model assets use separate persistent named volumes.
Keep volumes when replacing containers; back up data.
Frontend runs non-root; no public hosting, TLS, or auth.
Nginx proxies API paths; Vite is for native development.

04 Validation and remaining work

3 backend and 3 frontend unit tests passed; build passed.
Real samples produced text; long-form processing was checked.
Docker: health, all samples, duplicate, list, search, and errors passed. Replacement preserved records and audio/model hashes.
Manual listening-based accuracy review remains pending.
No word-error-rate, production-load, multi-replica, or exhaustive browser/accessibility claim is made by these checks.
The supplied assignment starts at section 2; scope assumes no additional requirements from an omitted section 1.

Implementation sources and editable artifacts

backend/app/{routes,storage,audio,transcription,database}.py; compose.yaml; backend/gunicorn.conf.py
frontend/src/; docs/api-contract.md; docs/containers.md; docs/testing.md; README.md
Editable diagram: docs/architecture.mmd | Exact PDF drawing source: docs/build_architecture.py